

Supplementary Material: Leakage-Controlled Concept Graph Construction and Cold-Start Diagnostic Protocol for Knowledge Tracing

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Abstract

This supplement provides per-dataset baseline and ablation tables, paired significance tests, full cold-start strata, DDR downstream scatter plot, sequence-autocorrelation statistics, operator definitions, ground-truth disagreement examples, extended ground-truth metrics with node Jaccard, and multi-benchmark schematic figures omitted from the main manuscript for length.

Supplementary index

Tables. S1–S5: per-dataset baseline diagnostics (Section 1); S6–S10: per-dataset graph ablation (Section 2); S11: paired significance tests on public benchmarks (Section 3); S12–S13: cold-start diagnostics (Section 8); S14: ground-truth metrics including node Jaccard (Section 7); S15: training parity (Section 11); S16: fold-level ΔAUC intervals (Section 12); S17: controlled leak injection (Section 13).

Sections and figures. Section S3: sequence-autocorrelation analysis (Section 4); Section S4: DDR augmentation operators (Section 5); Section S5: ground-truth disagreement examples (Section 6); Figure S1: multi-benchmark KT-graph schematics

Table 1: Diagnostic baseline results (train-only graphs). Baselines are used for protocol comparison only; no SOTA claim is made. Values are fold means with 95% bootstrap confidence intervals over valid+test sequence positions (`n_folds` as exported per dataset).

Model	AUC		ACC		NLL	
AKT	0.984	[0.982, 0.989]	0.951	[0.945, 0.962]	0.121	[0.093, 0.135]
BKT	0.780	[0.779, 0.781]	0.846	[0.845, 0.847]	0.379	[0.377, 0.380]
DGEKT	0.981	[0.981, 0.981]	0.943	[0.942, 0.943]	0.137	[0.136, 0.138]
DKT	0.984	[0.982, 0.989]	0.951	[0.945, 0.962]	0.120	[0.093, 0.134]
DyGKT	0.978	[0.978, 0.978]	0.934	[0.933, 0.934]	0.151	[0.151, 0.152]
GIKT	0.982	[0.982, 0.982]	0.946	[0.946, 0.946]	0.133	[0.132, 0.133]
<i>simpleKT</i>	0.984	[0.982, 0.989]	0.951	[0.945, 0.962]	0.121	[0.093, 0.135]
SKT	0.979	[0.976, 0.986]	0.938	[0.931, 0.954]	0.139	[0.107, 0.156]

Table 2: Diagnostic baselines: ASSISTments 2012 (three-fold learner CV (`n_folds=3`); train-only graphs).

Model	AUC		ACC		NLL	
AKT	0.968	[0.968, 0.969]	0.916	[0.914, 0.917]	0.175	[0.173, 0.176]
BKT	0.669	[0.667, 0.671]	0.713	[0.711, 0.715]	0.574	[0.573, 0.576]
DGEKT	0.960	[0.959, 0.961]	0.907	[0.906, 0.908]	0.186	[0.185, 0.188]
DKT	0.963	[0.961, 0.964]	0.910	[0.908, 0.911]	0.182	[0.180, 0.185]
DyGKT	0.952	[0.951, 0.952]	0.899	[0.898, 0.900]	0.200	[0.198, 0.203]
GIKT	0.967	[0.967, 0.968]	0.915	[0.914, 0.915]	0.175	[0.174, 0.177]
GKT	0.961	[0.960, 0.962]	0.907	[0.906, 0.908]	0.185	[0.184, 0.187]
<i>simpleKT</i>	0.968	[0.967, 0.968]	0.916	[0.915, 0.916]	0.173	[0.172, 0.175]
SKT	0.953	[0.952, 0.953]	0.898	[0.897, 0.899]	0.198	[0.196, 0.200]

(Figure 3); Figure S2: multi-benchmark DDR curves (Figure 4); Figure S3: DDR downstream scatter plot (Figure 2).

1 Per-dataset baseline diagnostics

Tables S1–S5 (Tables 1–5) reproduce the five diagnostic baseline tables (AUC, ACC, NLL with 95% bootstrap CIs) that the main manuscript summarises in its cross-dataset AUC summary table.

2 Per-dataset graph construction ablation

Tables S6–S10 (Table 6–10) report fold 0 train-only versus full-log metrics for each graph-augmented pyKT model. The main manuscript reports only the maximum $|\Delta\text{AUC}|$ and $|\Delta\text{ACC}|$ per dataset.

Table 3: Diagnostic baselines: XES3G5M (three-fold learner CV (`n_folds=3`); train-only graphs).

Model	AUC		ACC		NLL	
AKT	0.875	[0.874, 0.875]	0.858	[0.857, 0.858]	0.324	[0.323, 0.326]
BKT	0.636	[0.634, 0.638]	0.793	[0.791, 0.793]	0.490	[0.490, 0.492]
DGEKT	0.841	[0.841, 0.842]	0.844	[0.844, 0.845]	0.358	[0.356, 0.358]
DKT	0.858	[0.857, 0.858]	0.851	[0.851, 0.852]	0.342	[0.341, 0.342]
DyGKT	0.752	[0.752, 0.754]	0.813	[0.813, 0.814]	0.431	[0.429, 0.431]
GIKT	0.878	[0.878, 0.878]	0.860	[0.859, 0.860]	0.321	[0.320, 0.322]
GKT	0.834	[0.833, 0.835]	0.842	[0.841, 0.842]	0.364	[0.363, 0.365]
<i>simpleKT</i>	0.875	[0.874, 0.875]	0.858	[0.857, 0.859]	0.324	[0.323, 0.325]
SKT	0.625	[0.623, 0.626]	0.800	[0.799, 0.801]	0.482	[0.480, 0.483]

Table 4: Diagnostic baselines: Synthetic C2 (three-fold learner CV (`n_folds=3`); train-only graphs).

Model	AUC		ACC		NLL	
AKT	0.873	[0.870, 0.876]	0.808	[0.802, 0.812]	0.407	[0.401, 0.417]
BKT	0.835	[0.833, 0.838]	0.782	[0.780, 0.784]	0.460	[0.458, 0.462]
DGEKT	0.710	[0.634, 0.859]	0.724	[0.686, 0.794]	0.551	[0.432, 0.613]
DKT	0.862	[0.857, 0.868]	0.798	[0.788, 0.805]	0.421	[0.413, 0.434]
DyGKT	0.710	[0.631, 0.867]	0.726	[0.686, 0.801]	0.550	[0.421, 0.617]
GIKT	0.875	[0.871, 0.880]	0.809	[0.802, 0.815]	0.405	[0.398, 0.416]
GKT	0.873	[0.868, 0.878]	0.808	[0.799, 0.814]	0.407	[0.400, 0.420]
<i>simpleKT</i>	0.875	[0.871, 0.880]	0.809	[0.803, 0.814]	0.405	[0.399, 0.416]
SKT	0.708	[0.634, 0.853]	0.722	[0.686, 0.788]	0.554	[0.440, 0.613]

3 Paired significance tests (public benchmarks)

Table S11 (Table 11) reports paired t -test and Wilcoxon p -values for graph-augmented models versus *simpleKT* on the three public benchmarks. The main manuscript baseline-CV table lists only Wilcoxon p -values because three-fold CV limits Wilcoxon resolution ($p_{\min}=0.25$).

4 Sequence autocorrelation analysis

For every learner we order interactions by timestamp and, over consecutive within-learner pairs $(t-1, t)$, measure KC-repeat rate, item-repeat rate, label persistence, lag-one label autocorrelation ρ , and the AUC of the trivial previous-answer predictor (`scripts/compute_autocorrelation.py`).

5 DDR augmentation operator definitions

All operators act on the retained acyclic E_{pre} at perturbation strength $p \in \{0.05, 0.10, 0.20, 0.30\}$:

Table 5: Diagnostic baselines: Synthetic C5 (three-fold learner CV (`n_folds=3`); train-only graphs).

Model	AUC		ACC		NLL	
AKT	0.822	[0.819, 0.823]	0.750	[0.746, 0.752]	0.500	[0.496, 0.507]
BKT	0.791	[0.789, 0.794]	0.724	[0.722, 0.726]	0.534	[0.532, 0.536]
DGEKT	0.830	[0.827, 0.832]	0.756	[0.755, 0.757]	0.489	[0.486, 0.492]
DKT	0.826	[0.825, 0.827]	0.754	[0.752, 0.756]	0.493	[0.491, 0.496]
DyGKT	0.826	[0.824, 0.828]	0.751	[0.748, 0.753]	0.492	[0.490, 0.496]
GIKT	0.820	[0.817, 0.824]	0.749	[0.744, 0.754]	0.500	[0.496, 0.504]
GKT	0.810	[0.806, 0.814]	0.738	[0.730, 0.744]	0.526	[0.513, 0.538]
<i>simpleKT</i>	0.829	[0.828, 0.831]	0.755	[0.754, 0.757]	0.491	[0.489, 0.492]
SKT	0.810	[0.808, 0.813]	0.740	[0.737, 0.744]	0.511	[0.507, 0.513]

Table 6: Ablation contrasting *train-only* graph smoothing (protocol) with a *full-log* graph built from all learners before splitting (leaky construction). Graph-augmented rows are pyKT checkpoints consuming exported adjacency from P0. Δ is fold 0 full-log minus train-only. *to*: train-only graph; *fl*: full-log graph.

Model	AUC to	AUC fl	Δ AUC	ACC to	ACC fl	Δ ACC
GIKT	0.982	0.982	0.000	0.946	0.946	-0.000
SKT	0.979	0.976	-0.003	0.938	0.931	-0.008
DyGKT	0.978	0.978	-0.000	0.934	0.934	0.000
DGEKT	0.981	0.981	-0.000	0.943	0.943	-0.000

Table 7: Train-only vs. full-log graph ablation: ASSISTments 2012.

Model	AUC to	AUC fl	Δ AUC	ACC to	ACC fl	Δ ACC
GKT	0.961	0.961	-0.000	0.907	0.907	-0.000
GIKT	0.967	0.968	0.000	0.915	0.915	0.000
SKT	0.953	0.953	0.000	0.898	0.898	0.000
DyGKT	0.952	0.952	-0.000	0.899	0.899	-0.000
DGEKT	0.960	0.960	0.000	0.907	0.907	0.000

- **Attribute masking** masks a random subset of edge attributes without changing E_{pre} (DDR=0 by construction).
- **Edge dropping** removes each edge independently with probability p (expected $\text{DDR} \approx p$).
- **Subgraph sampling** performs a random walk until $\lceil (1-p)|\mathcal{C}| \rceil$ nodes are visited and returns the induced subgraph (boundary edges drop; internal connectivity preserved).
- **Node dropping** removes $\lceil p|\mathcal{C}| \rceil$ nodes and all incident edges.
- **prereq_preserve** draws the same per-edge budget as edge dropping but removes only transitively redundant edges, protecting the transitive-reduction backbone.

Table 8: Train-only vs. full-log graph ablation: XES3G5M.

Model	AUC to	AUC fl	Δ AUC	ACC to	ACC fl	Δ ACC
GKT	0.834	0.835	0.002	0.842	0.842	0.000
GIKT	0.878	0.878	0.000	0.860	0.860	0.000
SKT	0.625	0.624	-0.000	0.800	0.800	0.000
DyGKT	0.752	0.749	-0.003	0.813	0.812	-0.001
DGEKT	0.841	0.841	-0.000	0.844	0.844	-0.001

Table 9: Train-only vs. full-log graph ablation: Synthetic C2.

Model	AUC to	AUC fl	Δ AUC	ACC to	ACC fl	Δ ACC
GKT	0.873	0.873	-0.001	0.808	0.807	-0.001
GIKT	0.875	0.874	-0.000	0.809	0.809	0.000
SKT	0.708	0.708	0.000	0.722	0.723	0.001
DyGKT	0.710	0.710	0.000	0.726	0.726	-0.000
DGEKT	0.710	0.711	0.001	0.724	0.725	0.001

Table 10: Train-only vs. full-log graph ablation: Synthetic C5.

Model	AUC to	AUC fl	Δ AUC	ACC to	ACC fl	Δ ACC
GKT	0.810	0.812	0.002	0.738	0.738	0.000
GIKT	0.820	0.820	-0.000	0.749	0.747	-0.002
SKT	0.810	0.809	-0.001	0.740	0.739	-0.001
DyGKT	0.826	0.826	0.000	0.751	0.751	0.001
DGEKT	0.830	0.830	0.000	0.756	0.756	-0.001

6 Ground-truth disagreement examples (Junyi)

The cross-validation procedure exports qualitative disagreement cases to `results/tables/disagreement_examples.csv`. Inferred-only edges are predominantly bidirectional co-occurring arithmetic pairs (e.g. `number_line` $\leftrightarrow$ `addition_1`). Expert-only edges are systematic level-progression annotations (e.g. `equivalent_fractions_2` $\rightarrow$ `equivalent_fractions`). The `wrong_direction` category contains many reversed level-progression pairs such as `addition_1` $\rightarrow$ `addition_2`.

7 Ground-truth metrics with node Jaccard

Table S14 (Table 13) extends the main manuscript ground-truth table with node Jaccard across the same top- K cutoffs.

Table 11: Paired comparisons against *simpleKT* on public benchmarks (three-fold learner CV). Δ is mean AUC difference (model minus *simpleKT*); fold-level magnitudes in the main baseline-CV table are the primary evidence. p_t and p_W are reported for transparency only (underpowered at $n=3$ folds).

Dataset	Model	Δ AUC	p_t	p_W
ASSISTments 2012	GKT	-0.0069	<0.001	0.25
	GIKT	-0.0005	0.001	0.25
	SKT	-0.0154	<0.001	0.25
	DyGKT	-0.0160	<0.001	0.25
	DGEKT	-0.0079	<0.001	0.25
XES3G5M	GKT	-0.0411	<0.001	0.25
	GIKT	+0.0031	<0.001	0.25
	SKT	-0.2501	<0.001	0.25
	DyGKT	-0.1223	<0.001	0.25
	DGEKT	-0.0334	<0.001	0.25
Junyi Academy	GIKT	-0.0021	0.445	1.00
	SKT	-0.0055	0.031	0.25
	DyGKT	-0.0066	0.100	0.25
	DGEKT	-0.0034	0.267	0.25

Table 12: Sequence-autocorrelation diagnostics on the three public benchmarks. *KC rep.* and *Item rep.* are the fractions of consecutive within-learner interactions that repeat the previous KC or item; *Persist. acc.* is $P(y_t=y_{t-1})$; *Lag-1 ρ* is the lag-one label autocorrelation; *Prev-AUC* is the AUC of the trivial predictor $\hat{y}_t=y_{t-1}$.

Dataset	KC rep.	Item rep.	Persist. acc.	Lag-1 ρ	Prev-AUC
Junyi Academy	0.755	0.755	0.786	0.244	0.622
ASSISTments 2012	0.662	0.000	0.664	0.207	0.604
XES3G5M	0.211	0.145	0.786	0.342	0.671

8 Cold-start tables (full)

Table S12 (Table 14) reports fold 0 *simpleKT* metrics by KC-frequency stratum; Table S13 (Table 15) lists three-fold per-model AUC and $\Delta_{\max}$ by stratum. The main manuscript summarises both in a single table.

9 DDR downstream scatter plot

10 Additional schematic figures

11 Training parity

Table S15 (Table 16) outlines hyperparameter parity and model configuration across pyKT and native baselines. Settings shared by all models are kept out of the table: learning rate 0.001 (Adam), zero hyperparameter-search budget, split seed 42, identical

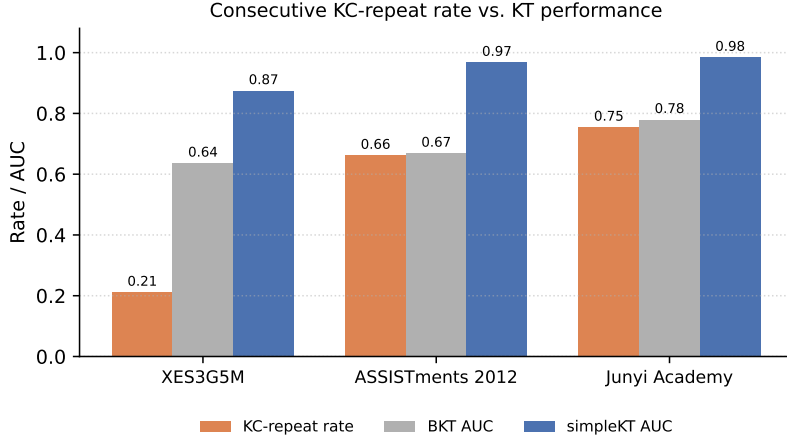


Fig. 1: Consecutive KC-repeat rate against *simpleKT* AUC across the three public benchmarks.

Table 13: Ground-truth cross-validation on Junyi: edge/direction/reachability metrics and node Jaccard across top- K cutoffs. The main manuscript table reports edge precision/recall, direction agreement, and reachability F1 only.

K	Edge F1	Dir. agr.	Rch. F1	Node Jaccard
200	0.071	0.346	0.008	0.290
500	0.109	0.225	0.021	0.453
800	0.150	0.280	0.062	0.572
$ \text{expert} =1131$	0.183	0.261	0.104	0.632
1200	0.185	0.262	0.126	0.641
1800	0.188	0.278	0.260	0.644
2500	0.187	0.285	0.339	0.610
5000 (cap)	0.187	0.285	0.339	0.610

processed splits under both train-only and full-log graph constructions, and checkpoint selection by best validation AUC with early stopping (BKT is fit by L-BFGS on NLL without early stopping). Epoch caps and batch sizes differ across families (30 epochs for sequence checkpoints vs. 10 for graph backbones), reflecting per-epoch cost; the main text flags this residual budget difference as a caveat on model-comparison claims. “Native” denotes our re-implementations in `src/models`; graph models consume the exported E_{pre} and, where available, E_{sim} edges.

12 Fold-level ΔAUC intervals for primary pairs

Table S16 (Table 17) reports the mean AUC difference (ΔAUC) between the primary model pairs on XES3G5M under train-only graph construction, with 95% paired- t intervals computed over the three learner-disjoint CV folds. With $n=3$ folds these

Table 14: Per-stratum cold-start KC diagnostic with *simpleKT* on fold 0. The n column counts test interactions whose KC falls in the stratum train-frequency bin; *out of range* marks test KCs outside the four canonical bins. Stratum mass varies by dataset split; under default bins Synthetic C2/C5 route all test interactions into the hot stratum (> 500 train-fold hits per knowledge component).

Dataset	Stratum	n	AUC	ACC	NLL
Junyi Academy	very_cold	19	1.000	0.684	0.456
	cold	256	0.447	0.621	0.633
	warm	11,257	0.615	0.727	0.572
	hot	7,725,588	0.718	0.831	0.413
	out of range	10	0.500	0.900	0.511
ASSIST'12	very_cold	49	0.608	0.755	0.631
	cold	707	0.769	0.717	0.565
	warm	9,017	0.765	0.687	0.587
	hot	807,178	0.693	0.704	0.574
XES3G5M	very_cold	161	0.810	0.814	0.424
	cold	2,592	0.798	0.794	0.453
	warm	17,655	0.790	0.805	0.431
	hot	1,902,424	0.717	0.796	0.467
	out of range	6	0.625	0.667	0.638
Synthetic C2	hot	60,000	0.624	0.683	0.607
Synthetic C5	hot	60,000	0.630	0.611	0.651

intervals are *descriptive* summaries of fold-to-fold variation, not substitutes for large-sample inference; we report them because the GKT deficit (-0.041) is more than an order of magnitude larger than the interval half-width (± 0.003), so the model-specific conclusion does not hinge on the test. Learner-level bootstrap intervals require full per-position prediction exports and can be reproduced with the released harness.

13 Controlled Leak Injection

Table S17 (Table 18) probes the sensitivity of the audit metrics by deliberately injecting 5% and 20% of test-fold interactions into the train-only prerequisite builder on XES3G5M (fold 0). Edge counts $|E_{\text{pre}}|$ are *candidate* edges before DAG pruning (the main-text DAG audit reports 701 retained edges after pruning on the clean fold). Candidate-edge count and TBMR_f increase monotonically with the injection rate, while $\text{ECR}_f^{\text{flag}}$ remains 0 (learner-id sets stay disjoint) and $\text{ECR}_f^{\text{overlap}}$ is already saturated at 1.0 on this transition-dense corpus. Flag-level checks alone therefore miss this contamination channel, supporting the tiered audit design in the main manuscript.

Table 15: Cold-start comparison of sequence-only (No-Graph) and graph-aware (Graph) KT models. Entries are three-fold mean AUC by KC train-frequency stratum; $\Delta_{\max}$ is the best Graph AUC minus the best No-Graph AUC in the same dataset/stratum.

Dataset	Stratum	No-Graph		Graph			
		DKT	SimpleKT	GKT	GIKT	DGEKT	$\Delta_{\max}$
assist2012	very_cold	0.723	0.724	0.695	0.726	0.724	+0.002
	cold	0.769	0.772	0.750	0.772	0.772	+0.000
	warm	0.751	0.756	0.724	0.757	0.756	+0.001
	hot	0.687	0.696	0.601	0.691	0.687	-0.005
junyi	very_cold	0.813	0.813	0.813	0.813	0.813	+0.000
	cold	0.505	0.505	0.505	0.505	0.505	+0.000
	warm	0.631	0.631	0.626	0.627	0.626	-0.004
	hot	0.719	0.719	0.704	0.713	0.711	-0.006
synthetic_c2	hot	0.619	0.626	0.554	0.634	0.634	+0.008
synthetic_c5	hot	0.628	0.630	0.524	0.633	0.633	+0.002
xes3g5m	very_cold	0.820	0.817	0.826	0.818	0.818	+0.005
	cold	0.802	0.804	0.793	0.804	0.803	-0.000
	warm	0.783	0.787	0.745	0.780	0.776	-0.007
	hot	0.708	0.717	0.606	0.709	0.703	-0.009

Table 16: Training parity and hyperparameters (Table S15)

Model	Codebase	Graph input	Epochs	Batch	ES patience	Runtime	Hardware
BKT	scipy L-BFGS	—	30	64	N/A	<5 min	CPU
DKT	pyKT stock	—	30	64	5	30–180 min	A100/V100
<i>simpleKT</i>	pyKT stock	—	30	64	5	30–180 min	A100/V100
AKT	pyKT stock	—	30	64	5	30–180 min	A100/V100
GKT	pyKT stock	$E_{\text{pre}} + E_{\text{sim}}$	10	4	5	30–180 min	A100/V100
GIKT	native	$E_{\text{pre}} + E_{\text{sim}}$	10	8	5	30–180 min	A100/V100
SKT	native	$E_{\text{pre}} + E_{\text{sim}}$	10	8	5	30–180 min	A100/V100
DyGKT	native	$E_{\text{pre}} + E_{\text{sim}}$	10	8	5	30–180 min	A100/V100
DGEKT	native	$E_{\text{pre}} + E_{\text{sim}}$	10	8	5	30–180 min	A100/V100

Table 17: Fold-level ΔAUC with paired- t 95% intervals, XES3G5M, train-only graphs (Table S16)

Model Pair	ΔAUC	95% CI (paired t)	p -value
GKT vs <i>simpleKT</i>	-0.041	[-0.044, -0.038]	<0.001
GIKT vs <i>simpleKT</i>	+0.003	[+0.003, +0.003]	<0.001

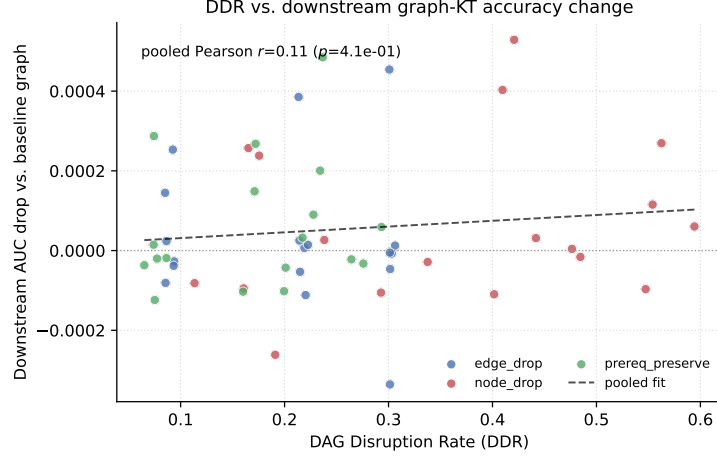


Fig. 2: Supplementary Figure S3. Downstream DGEKT sensitivity to prerequisite-graph disruption on ASSISTments 2012 and XES3G5M. Each point is one fold/operator/strength run; the vertical axis is AUC drop relative to the unperturbed train-only graph.

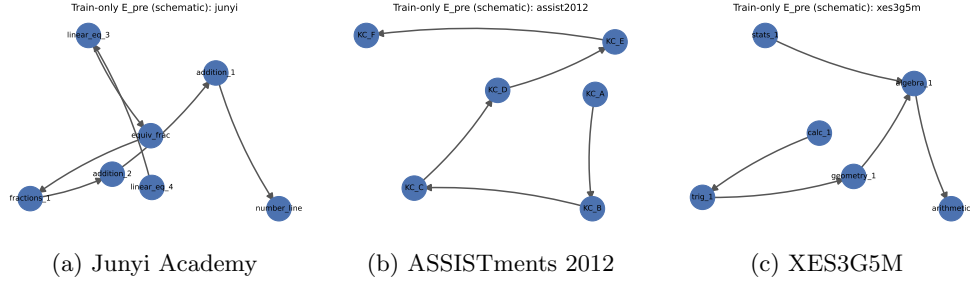


Fig. 3: Supplementary Figure S1. Schematic visualisations of high-support prerequisite chains in train-only E_{pre} after DAG audit (representative subgraphs). The main manuscript shows the Junyi panel only.

Table 18: Controlled Leak Injection, XES3G5M fold 0. $|E_{\text{pre}}|$ counts candidate edges before DAG pruning. (Table S17)

Injection rate	$ E_{\text{pre}} $	$\text{ECR}^{\text{overlap}}$	ECR^{flag}	TBMR
0%	1162	1.000	0.000	0.754
5%	1378	1.000	0.000	0.776
20%	1417	1.000	0.000	0.779

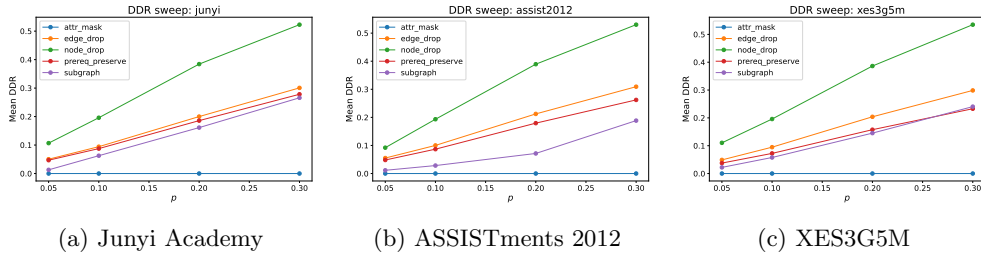


Fig. 4: Supplementary Figure S2. DDR versus perturbation strength p for five augmentation families across three benchmarks. The main manuscript shows the Junyi panel only.